Homework

DM3 · A family of curves · 45 min.

For every real number a, consider the function f_a defined by $f_a(x) = \frac{ax^2+1}{x^2+1}$. Let $\mathcal{C}_a$ its curve representative.

1. Determine the domain of f_a, then its limits at $-\infty$ and at $+\infty$.
2. Show that $f_a(x) - a = \frac{1-a}{x^2+1}$. Depending on the value of a, state the position of $\mathcal{C}_a$ in relation to the line with equation $y = a$.
3. Study the variations of f_a distinguishing the cases $a < 1$, $a = 1$ and $a > 1$. State any extrema and where they are attained.
4. Take $a = 3$. Solve, for $x \geq 0$, the inequality $|f_3(x) - 3| \leq 0{,}02$.